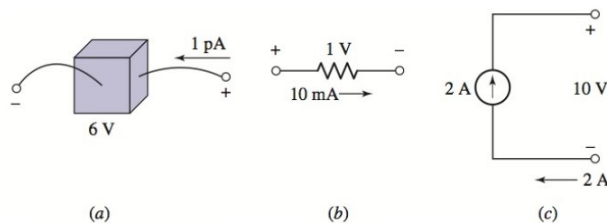


1. **Ohm's Law:** The following experimental data is acquired for an unmarked resistor, using a variable-voltage power supply and a current meter. Unfortunately, the current meter readout is somewhat unstable which introduces error into the measurement.

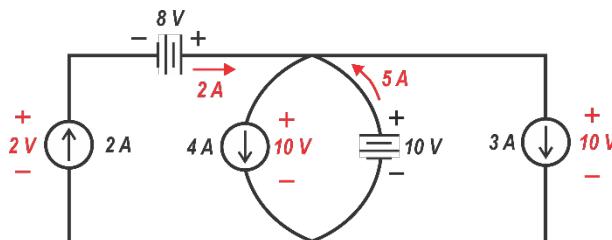
Voltage (V)	Current (mA)
-2.0	-0.89
-1.2	-0.47
0.0	0.01
1.0	0.44
1.5	0.70

- a) Using Microsoft Excel or other spreadsheet app, plot the measured current (x-axis) versus voltage (y-axis). Use a scatter-type plot to only display the x-y data points.
Submit a copy of your plot.
- b) Using a best-fit line (Trendline in Excel), **estimate the value of the resistance.** Ideally, the best fit line should cross (0,0). (Hint, include the equation for the best fit line. The slope of the equation is the resistance, don't forget to divide the number by 1×10^{-3} for the mA.)
2. **Ohm's Law:** An electric heater draws a current of 10 A from a 120 V source. What is the resistance of the heater?
3. **Power:** Determine the power **absorbed** by each of the following elements; (a) a block of resistive material, (b) a resistor shown with its schematic representation, and (c) an independent current source shown with its schematic representation

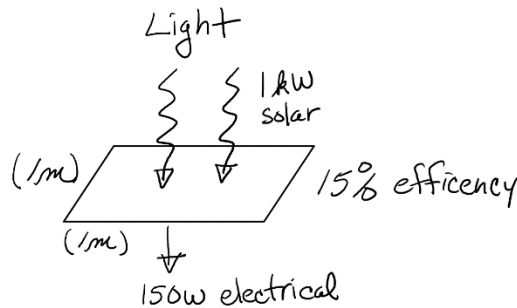


4. **Power:** For the following schematic diagram of independent sources, some sources are *delivering* power, and others are *absorbing* power. Determine which is which, and show that the algebraic sum of the power *delivered* is equal to the power *absorbed*.

Note: the values in red report one of the following two conditions: (1) the voltage across the independent current source or (2) the current flowing through the independent voltage source. Normally the values in red are not provided and are calculated during circuit analysis.

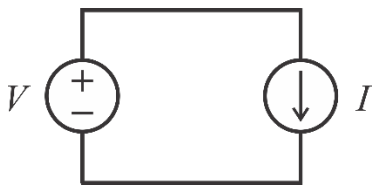


5. **Power and Efficiency:** An area of 1 m^2 will be able to capture 1 kW of solar power. A certain brand of solar panel is specified to convert solar energy to electrical energy with 15% efficiency.



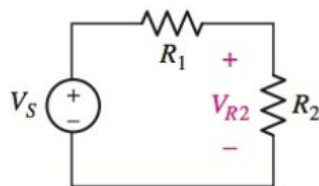
What is the panel area (in m^2) required to supply 1500 W of output power?

1. **Power:** For the following circuit, an independent voltage source is connected to an independent current source. Determine the power in the voltage source and whether the voltage source is **delivering** or **absorbing** power in each of the following cases.



- $V = 2 \text{ V}$ and $I = 4 \text{ A}$
- $V = 3 \text{ V}$ and $I = -2 \text{ A}$
- $V = -6 \text{ V}$ and $I = -8 \text{ A}$
- $V = -5 \text{ V}$ and $I = 1 \text{ A}$

2. **Ohm's Law:** For the following circuit and using the fact that the **total power supplied** by the voltage source must equal the **total power absorbed** by the two resistors, show that the equation is true.



$$V_{R_2} = V_s \frac{R_2}{R_1 + R_2}$$

Note: As this is a series connected circuit, then the same current flows through each element (a requirement for the conservation of charge). The above equation is also known as the "Voltage Divider" equation.